

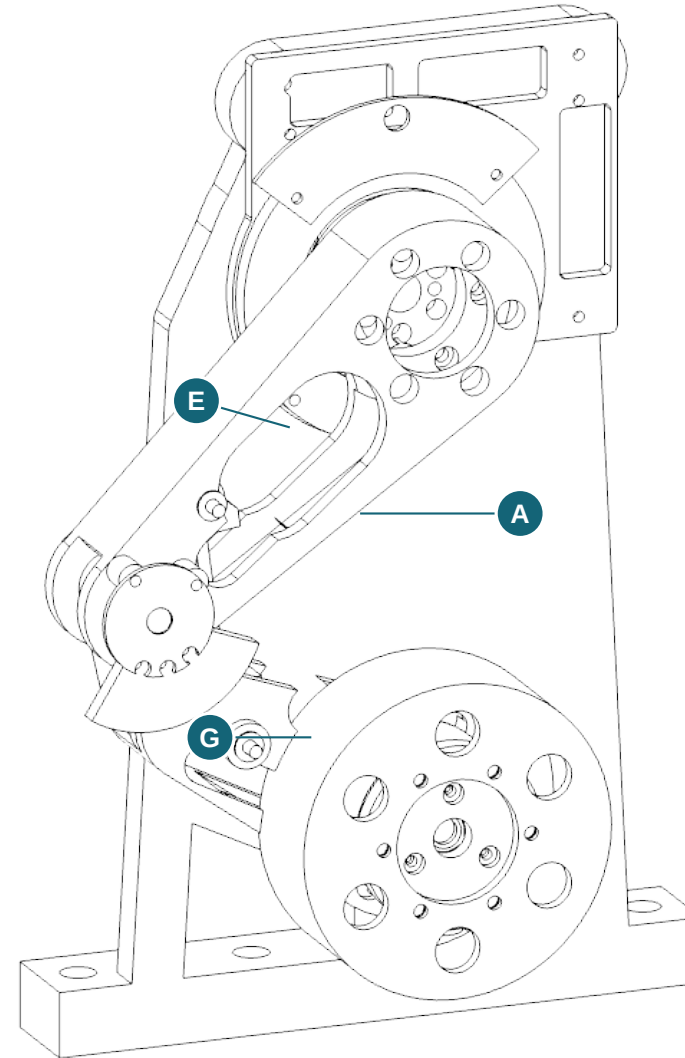
BENI / Single-leg assembly

Illustrated bench manual - current Amazon pins and revised ABS parts

BUILD IN THIS ORDER

- 01-04** Prepare loose parts
- 05-09** Build the shoulder
- 10-14** Build wheel end and knee
- 15-20** Fit cartridge and retainers
- 21** Hand-test the supported leg

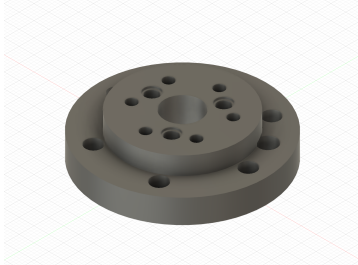
Both motors unplugged.
Stand clamped to the bench.
Wheel clear of bench and floor.
Keep one hand on the distal side.



Current CAD views, exploded only for illustration.

Identify your printed parts

Use the four revised parts. Match the remaining parts to your earlier batch.



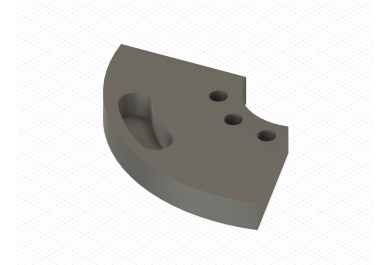
D NEW shoulder hub



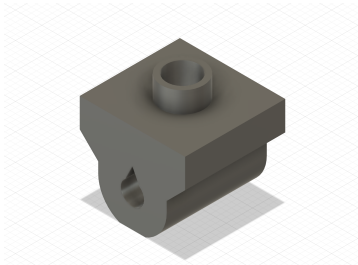
E NEW proximal link



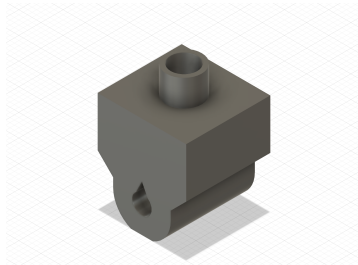
G NEW distal link



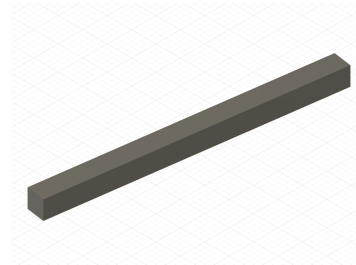
K NEW stop plate



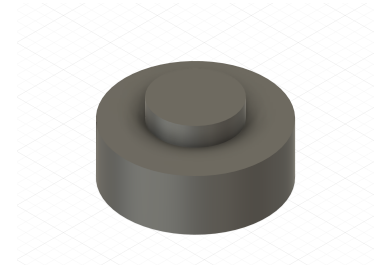
L Upper cartridge eye



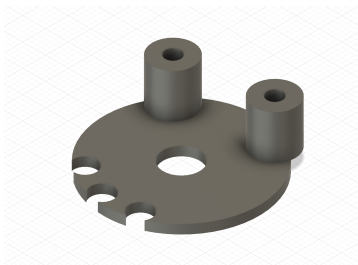
N Lower cartridge eye



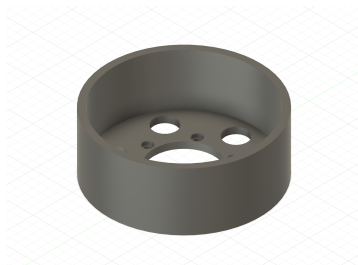
M Guide bar



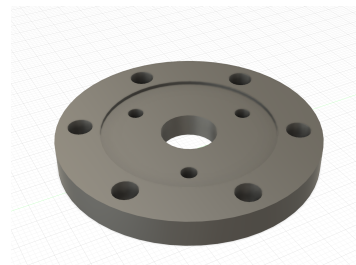
Q Knee-pin spacer



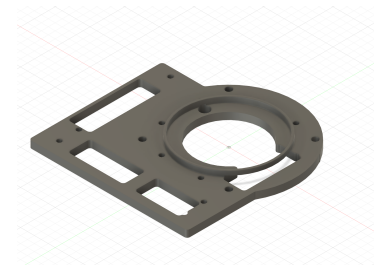
P Keeper bracket



R No-tyre shell



J Wheel hub



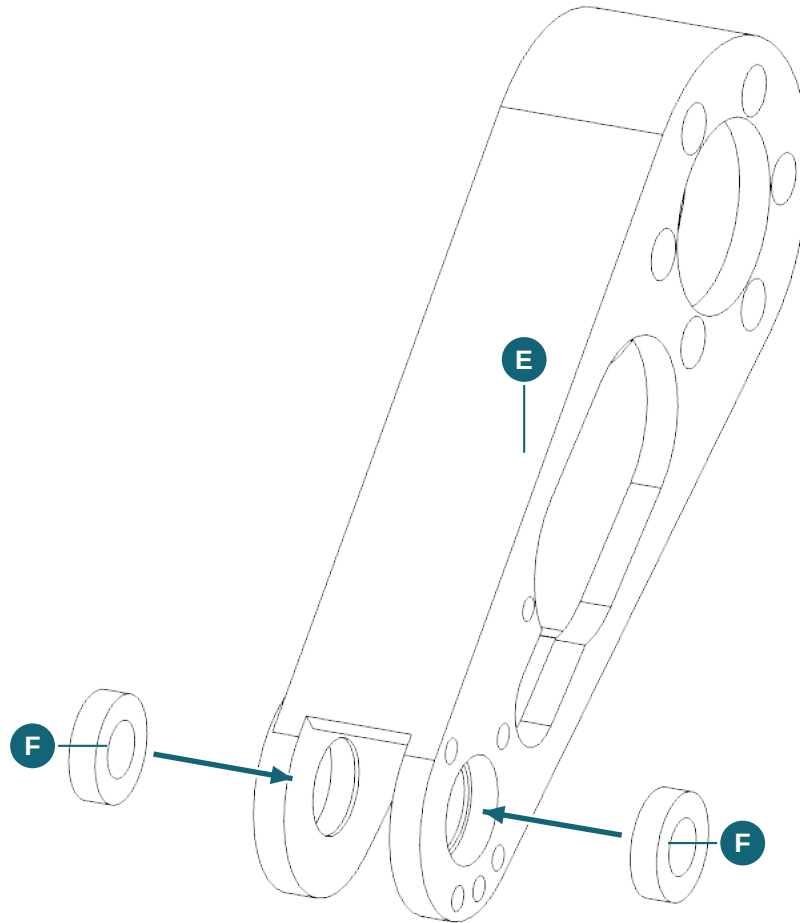
B Shoulder plate

Also use the current stand (A), cable cover and front post. The stand and plate need the released Ø4.5 M3 receivers. Check their print-file revision before fitting inserts.

Inboard = toward the stand/body. Outboard = away from the stand. C = GIM6010 shoulder motor; H = GIM4305 wheel motor; F = 6800 bearing.

01 Fit the two knee bearings

Work with the loose proximal link before installing inserts.



HARDWARE



2 x 6800

Use undamaged bearings.



1 x D10 x 35

Bought steel knee pin.

IN THIS VIEW



New proximal link



6800 bearings

CHECK

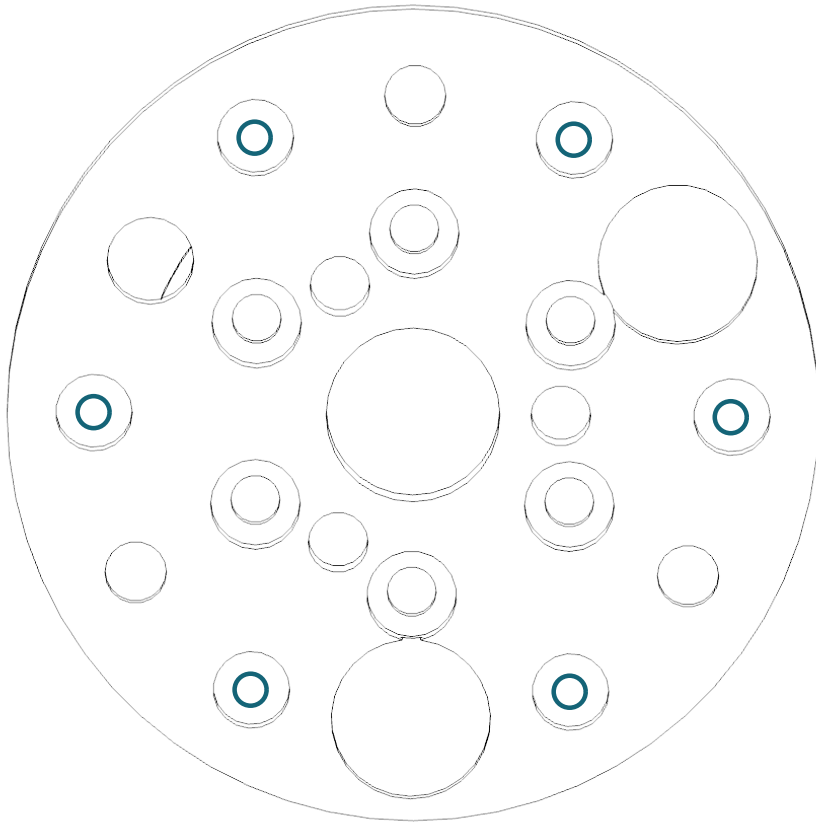
Both bearings sit square, without rock. Pass the steel pin through each bearing separately.

Press only on the outer bearing race; transfer old bearings only if they come out undamaged.

02 Insert the new hub and proximal link

Circled holes take inserts. Let each insert cool before starting a screw.

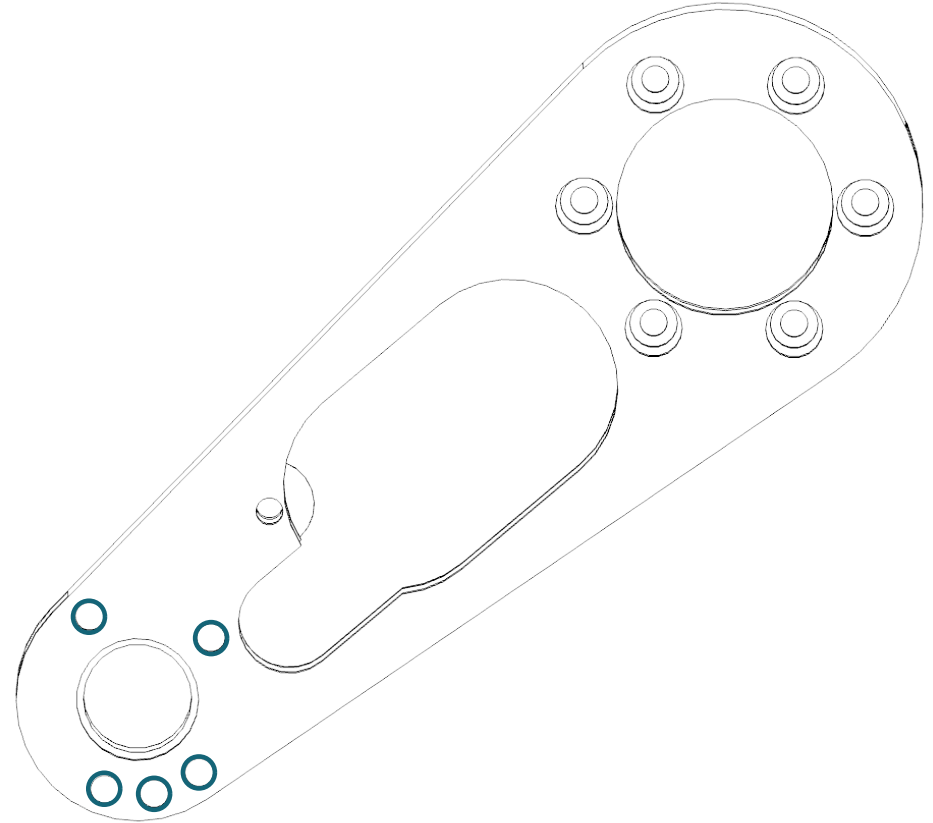
D / SHOULDER HUB



6 x M4 x 8 inserts

Enter from the outboard / link face.

E / PROXIMAL LINK



5 x M3 inserts

3 stop-plate holes + 2 keeper holes.

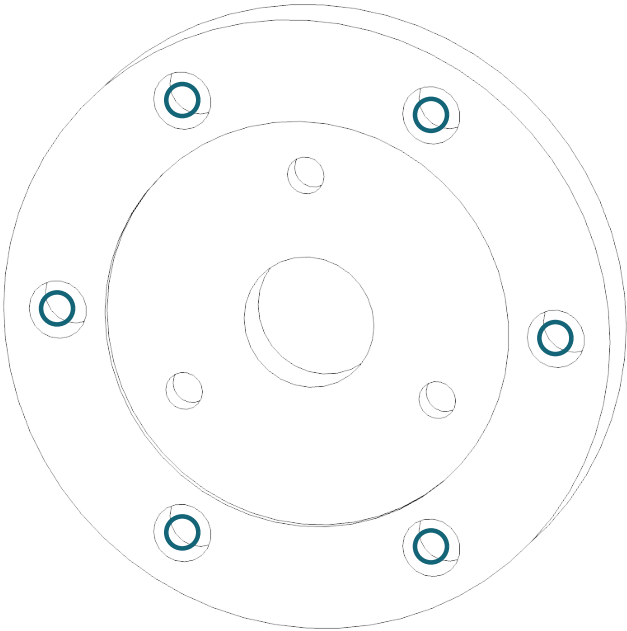
CHECK Inserts are flush, straight and firm after cooling. Keep heat away from the root-dowel sockets.

The distal link, stop plate and cartridge eyes take no heat-set inserts.

03 Finish the exposed insert receivers

Complete these before closing the motor and fixture joints.

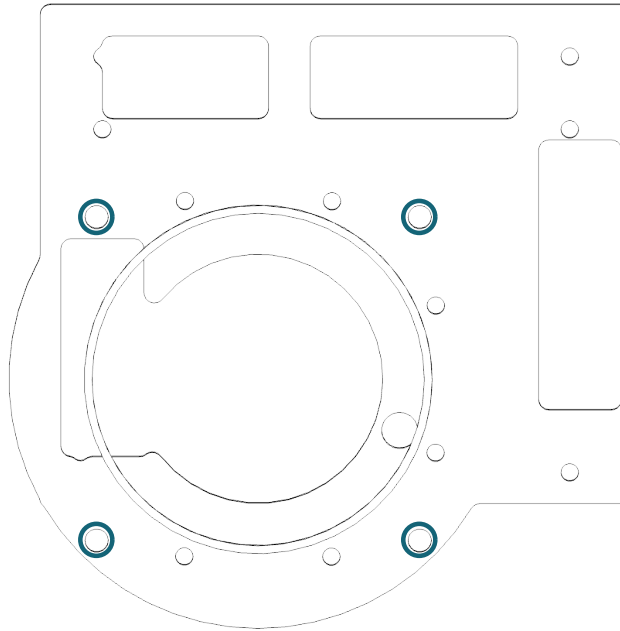
J / WHEEL HUB



6 x M4 x 8

Enter from motor face.

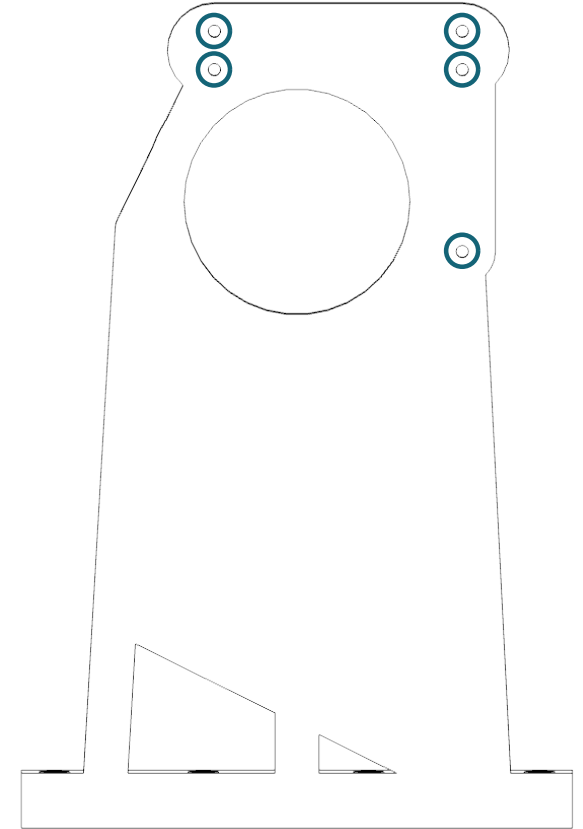
B / SHOULDER PLATE



4 x M3

Enter from outboard face.

A / STAND



5 x M3

Enter from plate-mount face.

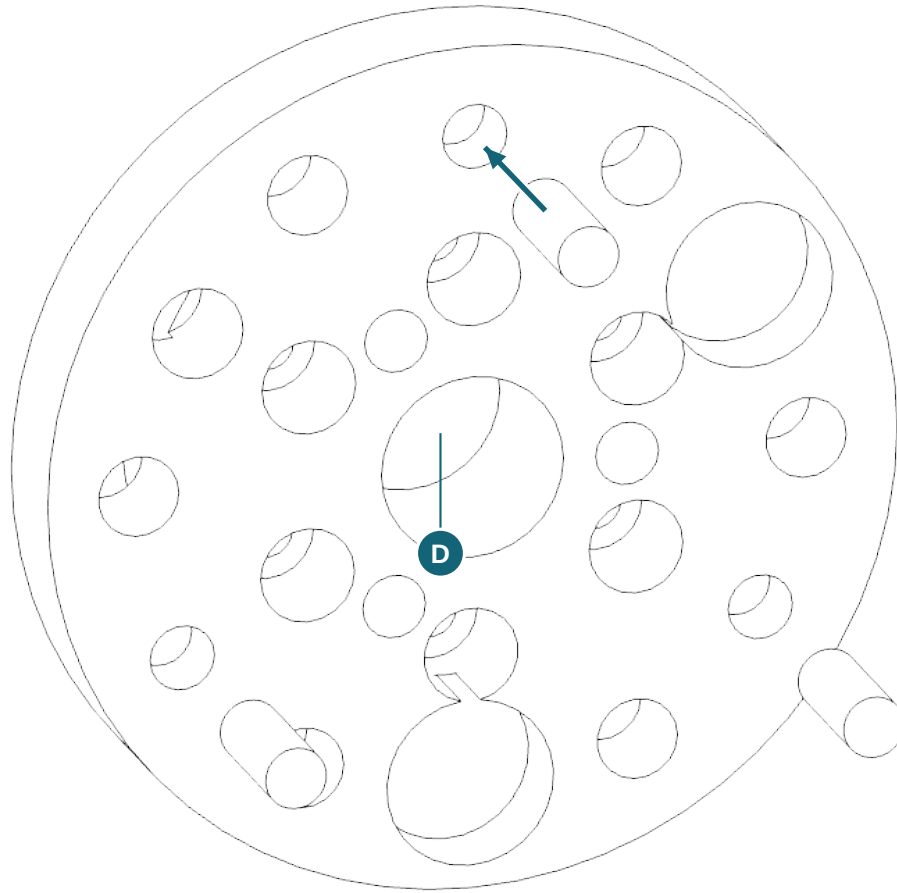
CHECK

Wheel-hub inserts project 2.0 mm on the rim side; shoulder-plate inserts finish flush.

Use a depth-controlled tip. The stand has 6.0 mm pockets for the approved 5 mm M3 inserts.

04 Seat the three root dowels

Support the detached hub flat. Keep the pin axes straight.



HARDWARE

 3 x Ø4 x 10

Bought root dowels.

IN THIS VIEW

D New shoulder hub

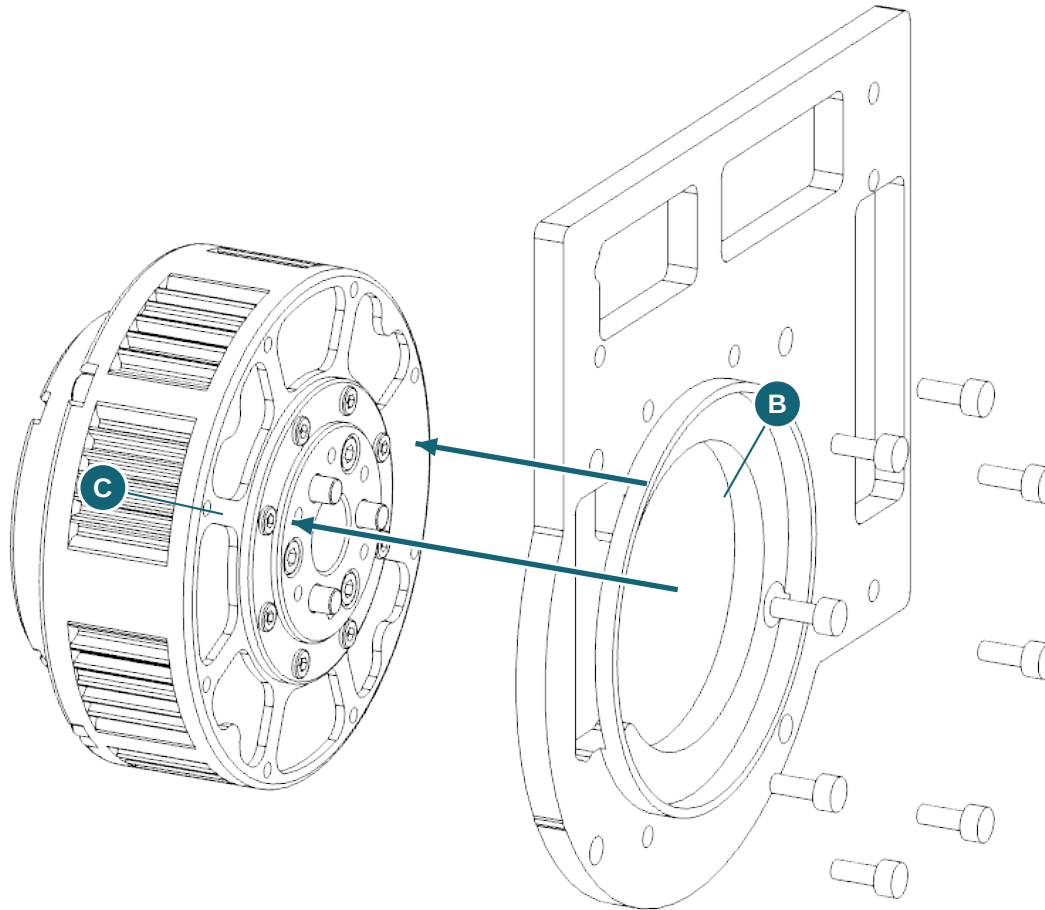
Stop for whitening, splitting or a cocked pin.

CHECK Seat each dowel 5.0 mm in its blind socket; half of the 10 mm pin stays exposed.

Trial-fit the proximal root by hand now, then remove it for shoulder assembly.

05 Slide the plate over the bare rotor

View from outboard. The housing stays behind the plate.



HARDWARE

 8 x M3 x 8

Shoulder housing screws.

IN THIS VIEW

C GIM6010 shoulder motor

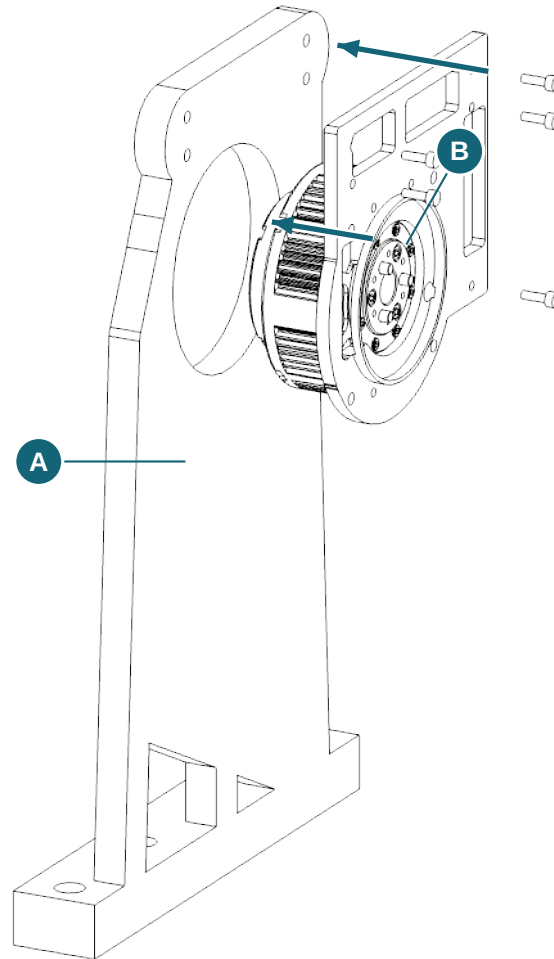
B Shoulder plate

CHECK Flat plate face against the stationary housing; cable-spiral lip faces outward.

Use M3 x 8 here. M3 x 10 bottoms before this plate is clamped.

06 Fasten the shoulder to the stand

Clamp or bolt the stand to the bench before adding the leg.



HARDWARE

 5 x M3 x 10

Plate-to-stand screws.

IN THIS VIEW

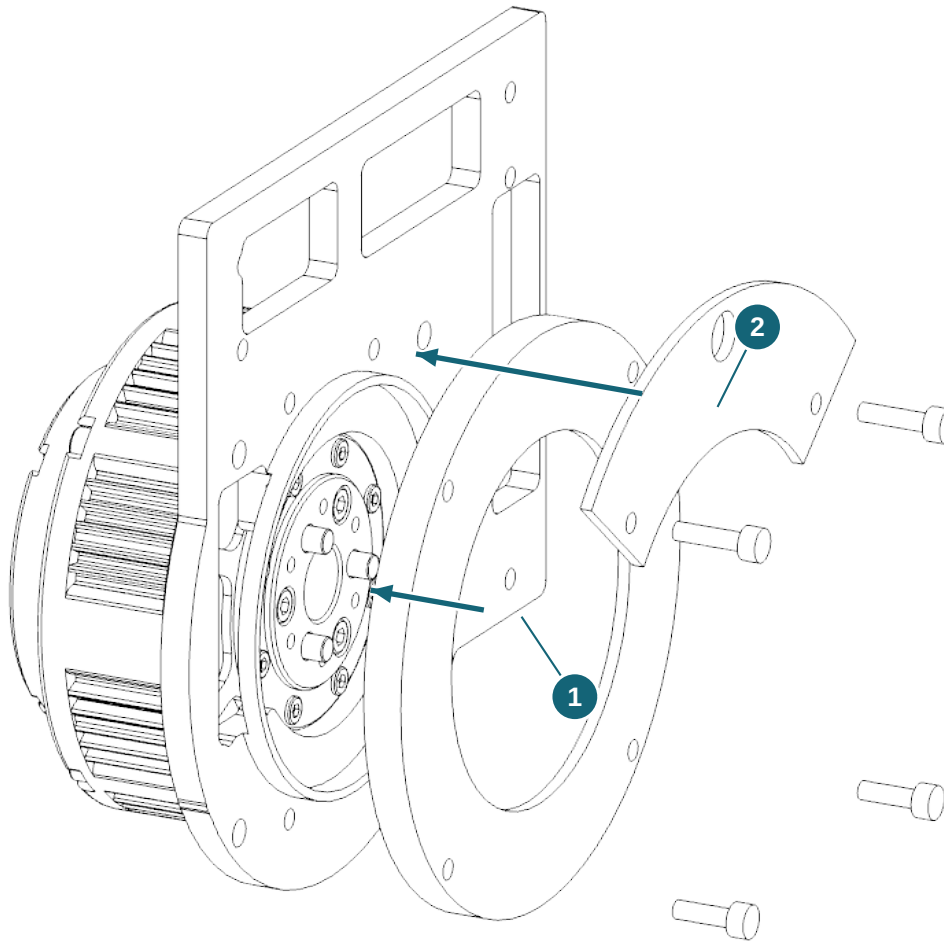
- A** Clamped stand
- B** Plate + motor from step 05

CHECK The five plate holes line up with the stand inserts; all screws finger-start.

Keep the motor power and communication leads unplugged.

07 Route the cable; fit cover and post

Complete this area while you can still reach it.



HARDWARE

 2 x M3 x 12

Upper post + cover positions.

 2 x M3 x 10

Lower cover positions.

IN THIS VIEW

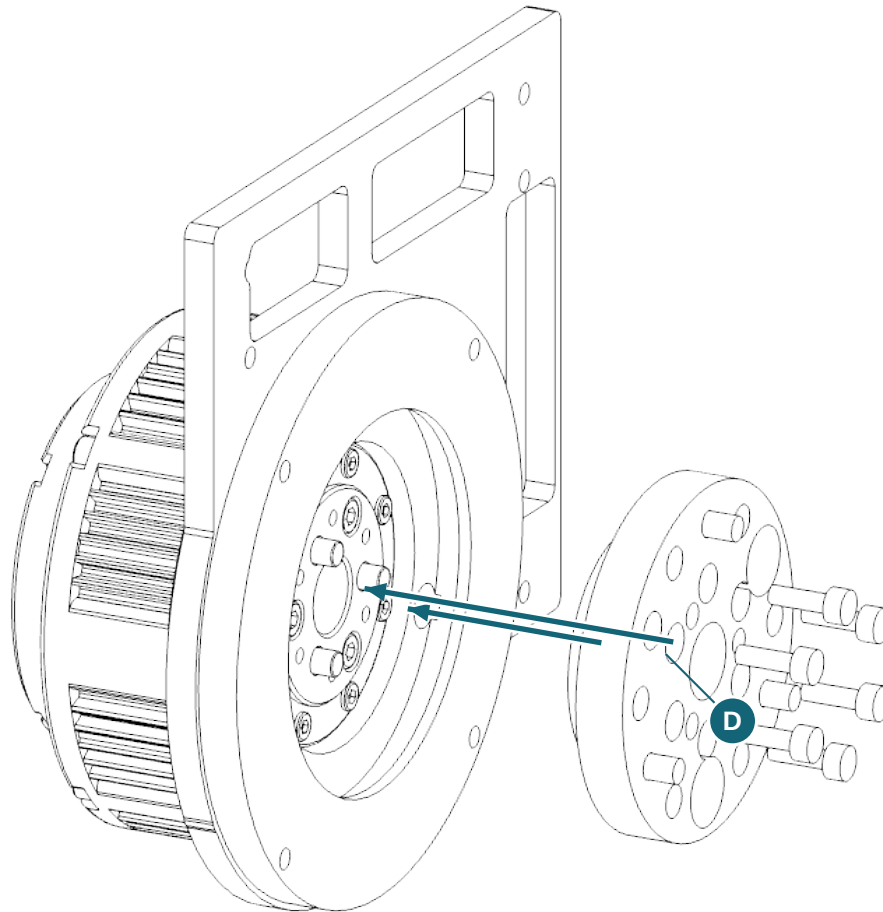
B Stationary shoulder plate

CHECK 1: Cover closes the cable spiral. 2: Front post sits outside the cover.

Route the real harness first. The printed cover has clearance holes and takes no inserts.

08 Fit the rotating shoulder hub

The motor factory pins enter the back of the new hub.



HARDWARE

 6 x M3 x 10

Hub-to-motor output screws.

IN THIS VIEW

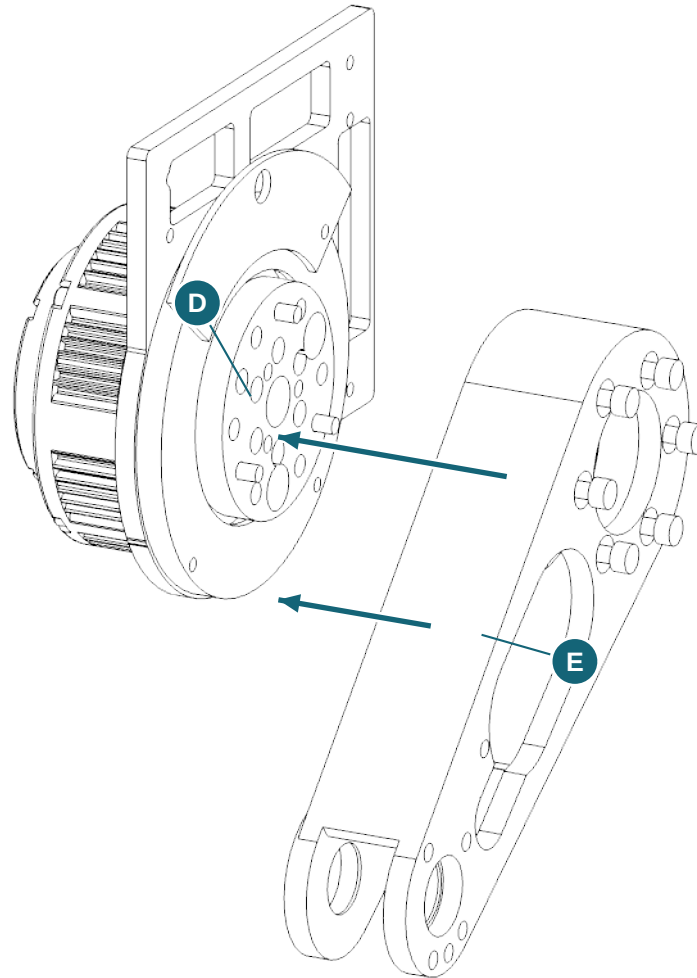
 D New hub with root dowels

CHECK Hub reaches the metal output face by hand; the three bought dowels point toward the link.

The three motor factory pins and the three bought root dowels are separate interfaces.

09 Attach the proximal root

Support the far end of the link throughout this step.



HARDWARE

 6 x M4 x 10

Link-to-hub clamp screws.

IN THIS VIEW

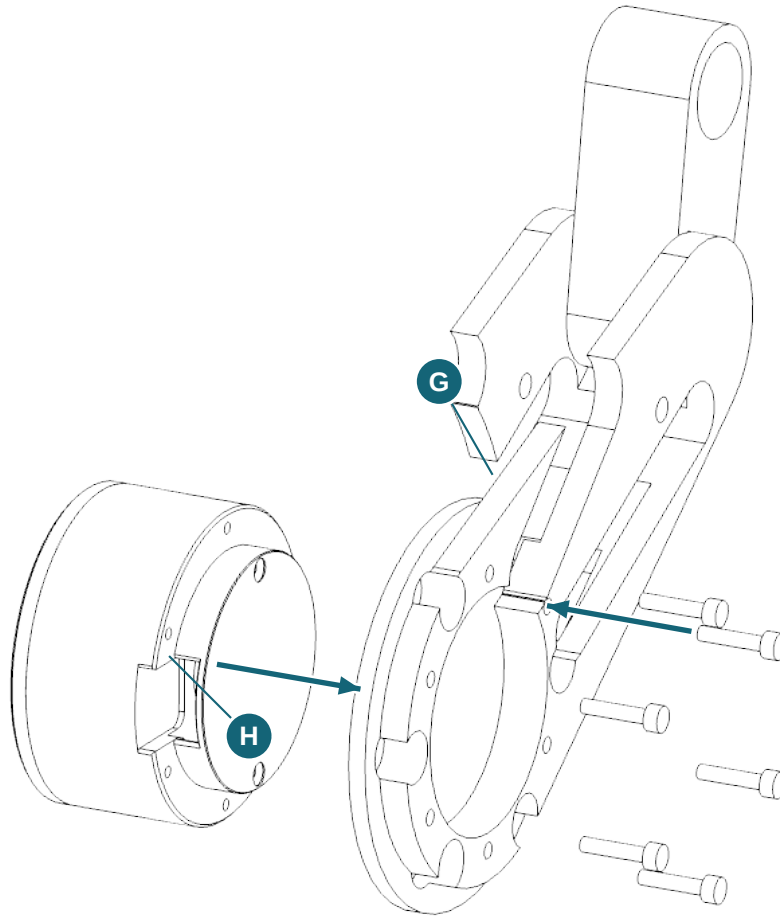
- D** Rotating shoulder hub
- E** New proximal link

CHECK Faces meet on all three dowels before tightening. Every screw head sits flat.

The six screws clamp the joint; do not use them to draw an incompatible fit together.

10 Mount the wheel motor

Detached distal link. View from inboard to show screw entry.



HARDWARE

 6 x M2.5 x 12

Enter from the inboard side.

IN THIS VIEW

G New distal link

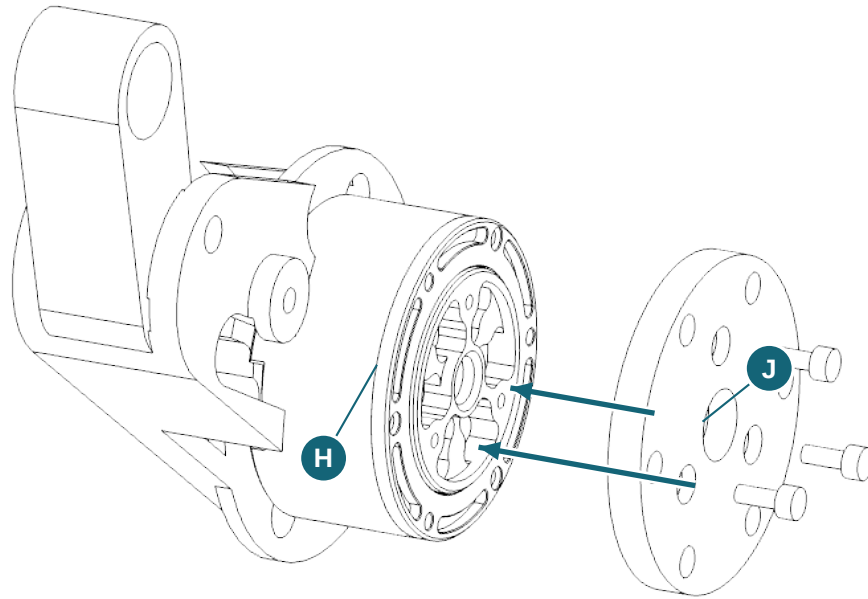
H GIM4305 wheel motor

CHECK The motor seats on the distal wheel ring without screw pull-down.

Keep the motor unplugged; support the loose assembly on the bench.

11 Fit the wheel output hub

View from outboard. The rim stays off for now.



HARDWARE

 3 x M3 x 8

Wheel hub to motor output.

IN THIS VIEW

H Wheel motor

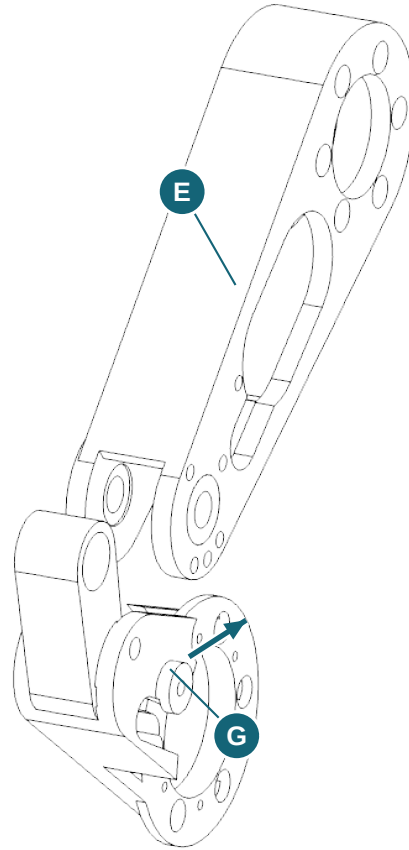
J Prepared wheel hub

CHECK Hub seats by hand and all three output screws start freely.

The six rim inserts were installed from the motor face in step 03.

12 Bring the distal receiver into the fork

The drawing isolates the links; support the fitted wheel motor and hub.



IN THIS VIEW

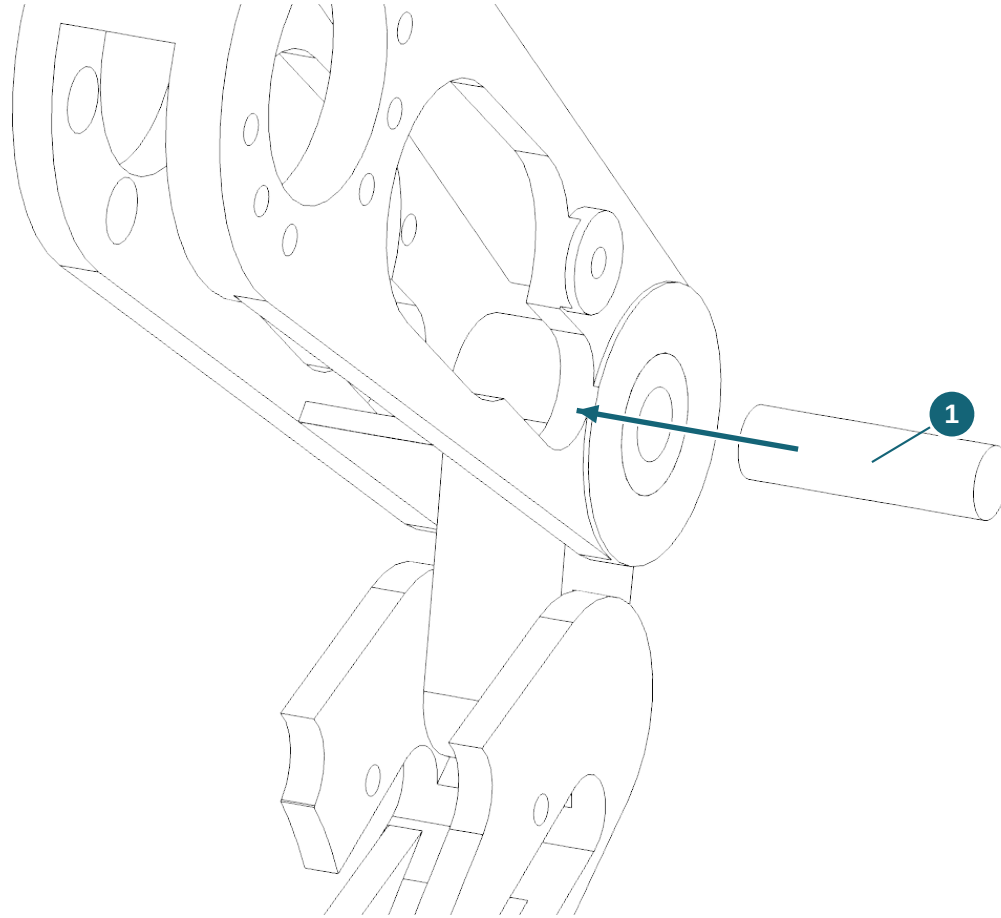
- E** Proximal fork + bearings
- G** Distal knee receiver

CHECK The distal receiver sits in the gap between the two bearings; all three pin bores align.

Slide in from the open side along the illustrated path. Keep the spring off.

13 Insert the steel knee pin

View from inboard. The pin travels through bearing - receiver - bearing.



HARDWARE

 **1 x D10 x 35**

Bought steel knee pin.

IN THIS VIEW

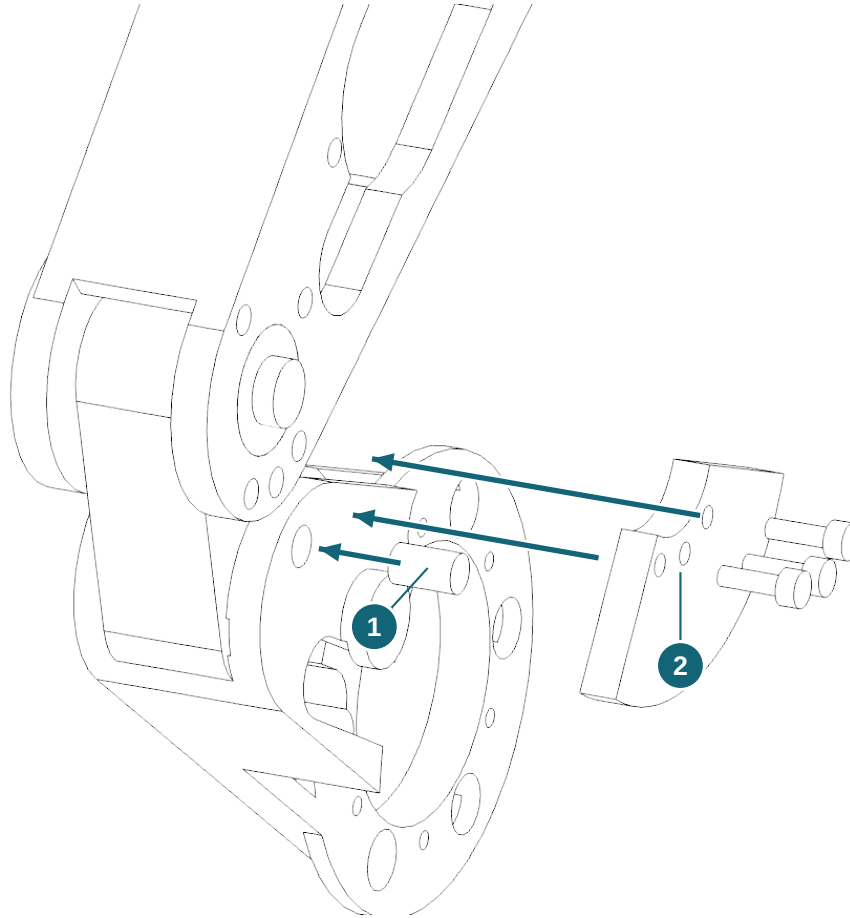
- E** Proximal fork
- G** Distal receiver

CHECK Firm-thumb insertion, full seating and hand withdrawal; no free spin or radial rock in the receiver.

Prove withdrawal now, before adding the stop, cartridge or keeper. Stop if the pin seizes.

14 Capture the stop dowel

Fit the dowel first, then close it in with the new stop plate.



HARDWARE

 1 x Ø6 x 10

Distal blind stop socket.

 3 x M3 x 10

Stop plate to proximal inserts.

IN THIS VIEW

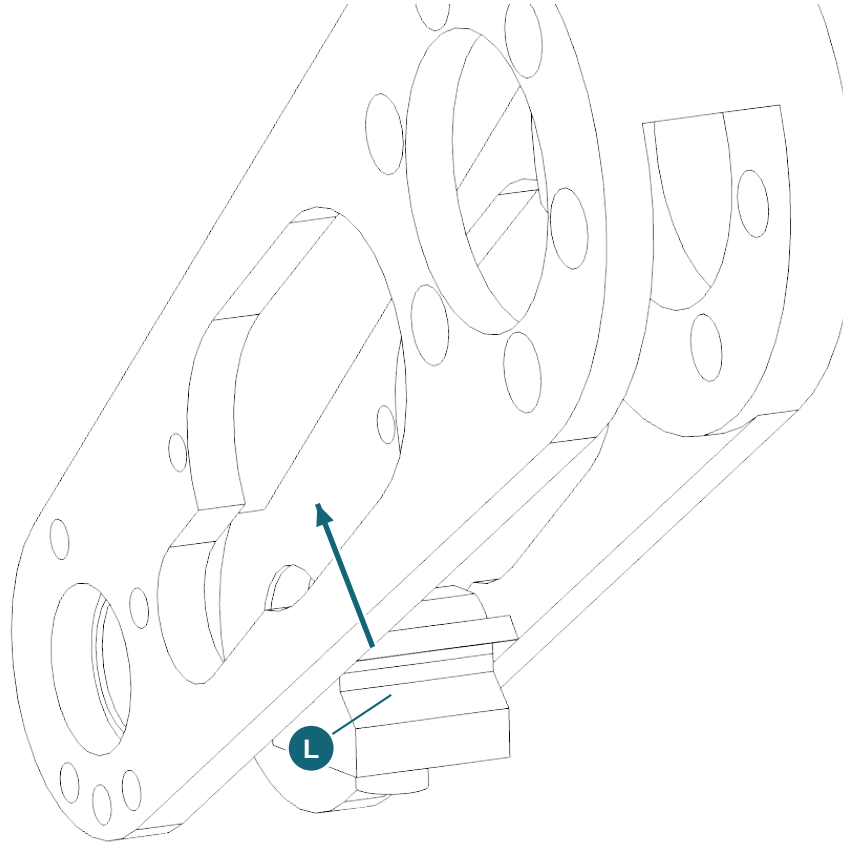
 K New closed-skin stop plate

CHECK Channel faces the dowel; closed skin faces outboard. Prove both stops with the spring absent.

The plate bolts to the proximal link. Allowed knee range: -8° to +15°. No glue or press fit here.

15 Place the upper cartridge eye

Keep the supported knee at its -8° extension stop.



IN THIS VIEW

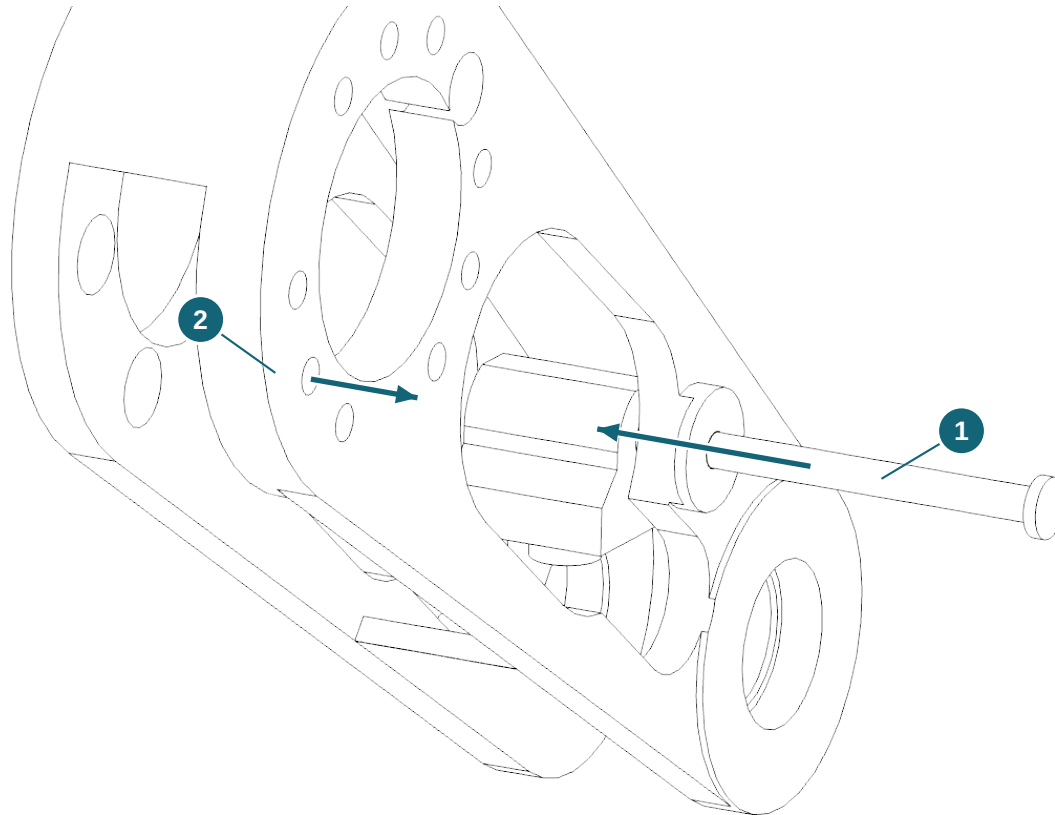
- E** Proximal upper clevis
- L** Upper cartridge eye

CHECK The side pivot passage aligns with the clevis. The raised round spring pilot faces the lower eye.

The round pilot hole carries the guide; the pointed-roof side hole carries the clevis pin.

16 Pin the upper eye

View from inboard. Add the washer and supplied cotter on the outboard end.



HARDWARE

 **1 x M4 x 40**

Head stays inboard.

 **1 x M4 washer**

ISO 7089 steel washer.

IN THIS VIEW

 Upper cartridge eye

Retention order: pin head /
printed joint / steel washer /
supplied cotter.

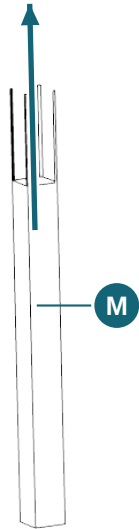
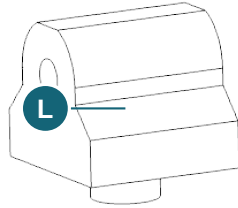
CHECK Pin passes by hand. Washer rests on the printed outboard face; supplied cotter is fully seated.

Orient the cotter radially away from the knee. No loose printed spacers.

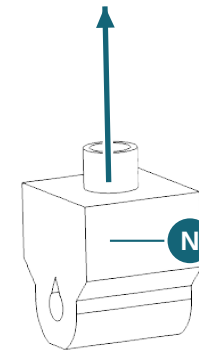
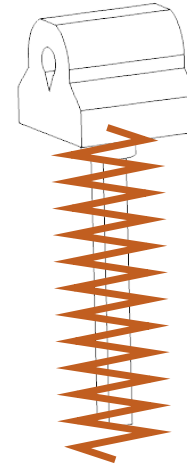
17 Guide first, then spring

Cartridge close-up with the surrounding links hidden. Keep the knee at -8°.

1 / GUIDE INTO ROUND PILOT HOLE



2 / SPRING OVER GUIDE; LOWER EYE FOLLOWS

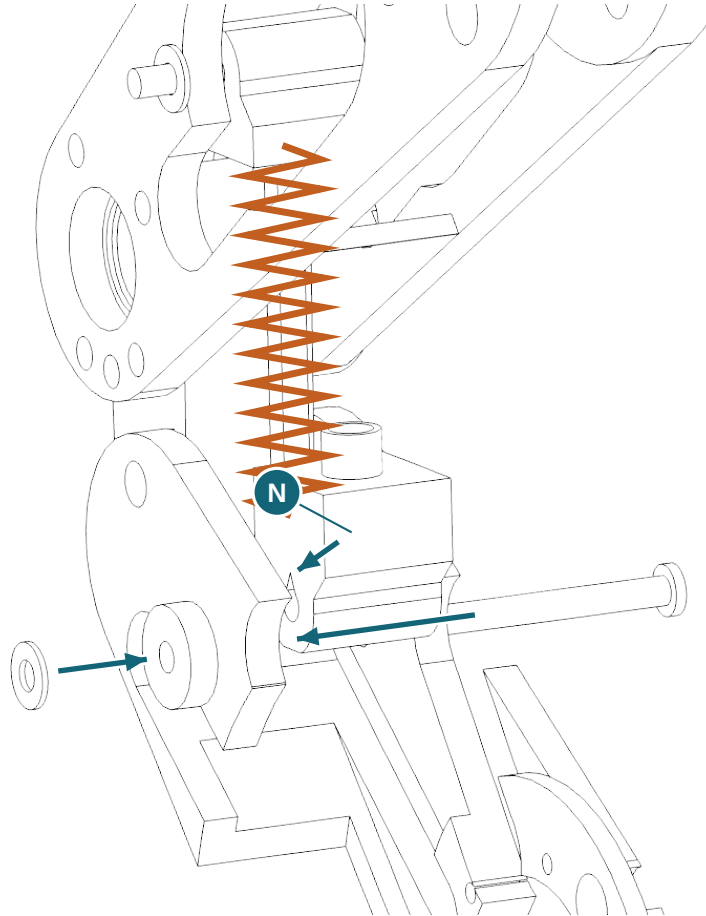


CHECK Both raised pilots face the spring. The lower eye slides freely on the guide.

Use the owned OD18 / ID9 / 50 mm spring. Orange spring symbol is schematic.

18 Pin the lower cartridge eye

Bring the lower eye into the distal clevis with the knee still at -8°.



HARDWARE

 **1 x M4 x 40**

Enter inboard to outboard.

 **1 x M4 washer**

Then the supplied cotter.

IN THIS VIEW

 **G** Distal lower clevis

 **N** Lower cartridge eye

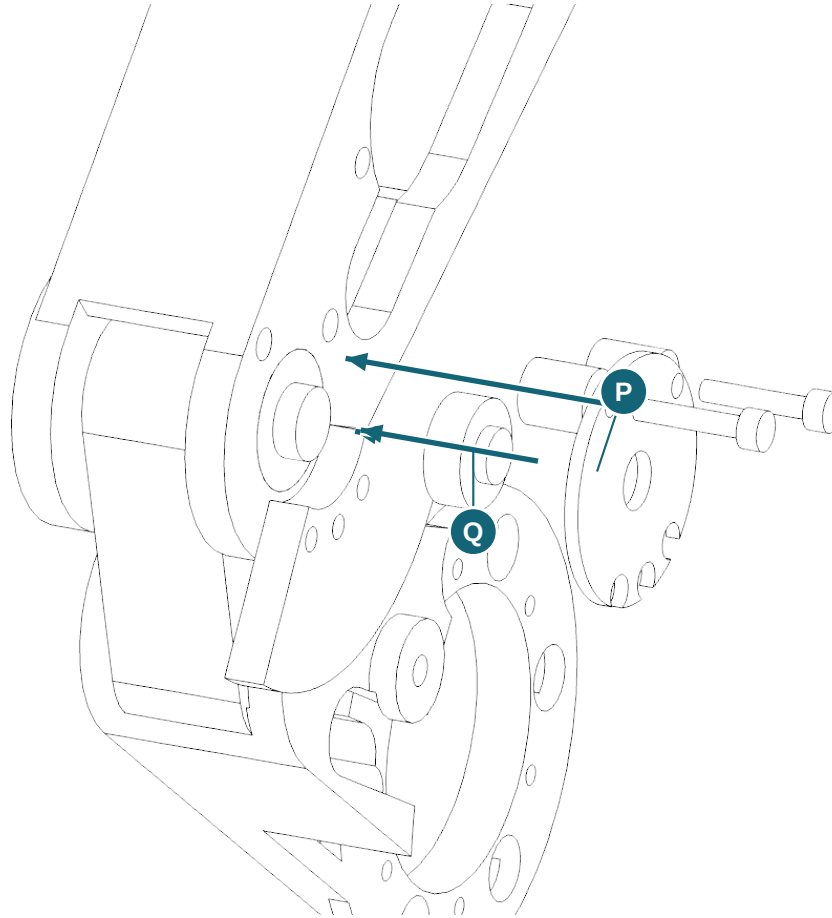
Turn the cotter radially away from the knee. Seat it fully without reshaping it.

CHECK The lower pin passes with fingertip pressure. Washer and cotter are on the outboard end.

Stop if more than slight hand compression is needed; do not pull the eye in with a pin or clamp.

19 Fit the outboard knee-pin keeper

The cartridge is hidden in this close-up for visibility.



HARDWARE

 2 x M3 x 16

Bracket to proximal inserts.

IN THIS VIEW

 Knee-pin spacer

 Keeper bracket

CHECK The spacer locator enters the bracket center hole; the bracket sits on its two mounting positions.

This limits outboard travel only. Keep hand control of the distal side to prevent inboard escape.

20 Attach the no-tyre shell

The wheel assembly remains suspended throughout this test.

HARDWARE

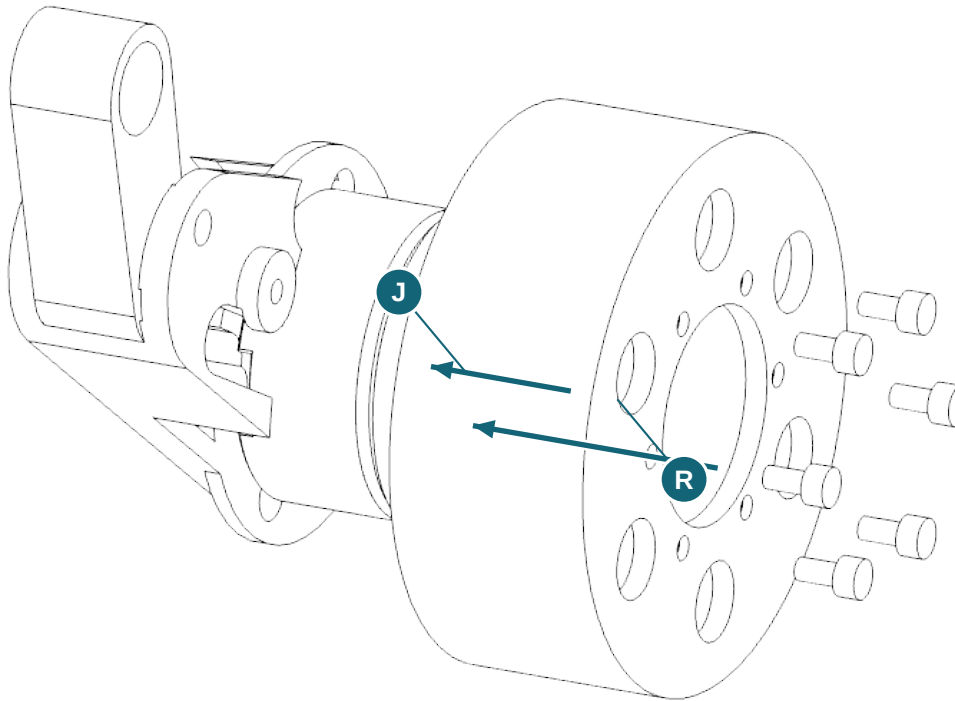
 6 x M4 x 8

Shell to wheel-hub inserts.

IN THIS VIEW

J Wheel output hub

R No-tyre test shell



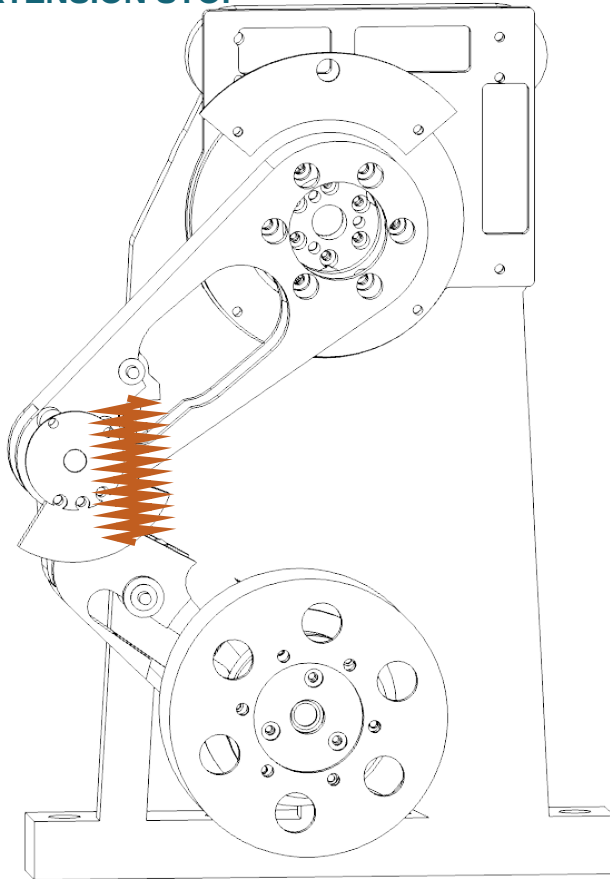
CHECK Shell seats without force; all six screws start freely. Leave the TPU tyre off.

Do not place the test shell on the bench or floor.

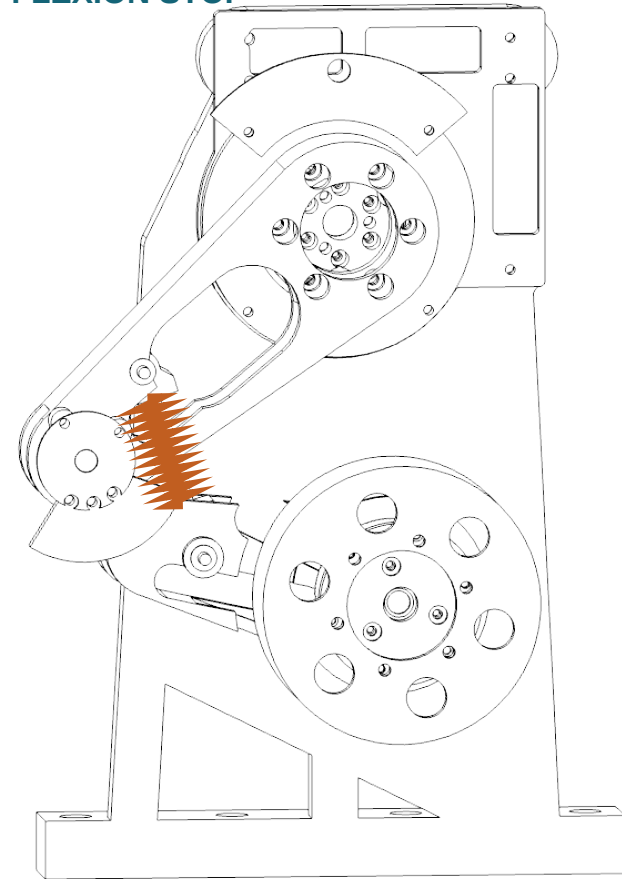
21 Check motion by hand

Stand clamped. Both motors unplugged. Wheel clear. One hand contains the distal side.

-8° / EXTENSION STOP



+15° / FLEXION STOP



-8° 0° +5° +10° +15°

CHECK Move slowly between the stops. Observe smooth movement and the settle / return tendency.

Stop for cracks, binding, coil contact, guide escape, moving pins, loose cotters or stop bypass.